BUSINESS PLANS BY STEPHEN CALHOUN

The Calhoun Forecasts

In 2001 and again in 2013, Stephen Calhoun wrote business plans describing technology that most people didn't have yet. Each prediction below is quoted *in his own words*, then checked against when it actually became widely available in the United States.

July 30, 2001 The CES plan **June 2013** The GenZPlay plan **June 2025** SASKI begins Twelve years apart, each time

NEW WIRING **Combined parts that already existed.** These tended to arrive fastest.

NEW INVENTION **Needed something nobody had built yet.** These took the longest, up to 18 years.

MARKET CALL **A forecast about people, markets or timing** rather than a product.

The CES plan

WRITTEN JULY 30, 2001 · PAGE NUMBERS REFER TO THE ORIGINAL PLAN

Quotes keep the original 2001 spelling and punctuation. Dates mark broad consumer availability in the US.

Strong hits

7 PREDICTIONS

WHAT YOU WROTE IN 2001	WHAT IT BECAME	YOUR EXACT WORDS	WHEN IT CAME TO FRUITION	TYPE
Microsoft putting Word and Excel on the Internet for a **monthly usage fee**, with no downloads and updates available immediately p.3	Office 365 launched in 2011 and is now Microsoft 365. You also described the whole subscription software model in general terms.	*"Microsoft has recently moved down the road towards putting applications (i.e. Word, Excel) on the Internet to be accessed, instead of installed inside of everybody's PC... What better way to sell somebody a product then directly thru the Internet with a monthly usage fee? Users would not have to download the latest updates or wait days for their new software to arrive in the mail. The latest and greatest business and recreational software will be available immediately for our use."*	**June 2011** (Office 365 launch). Renamed Microsoft 365 in April 2020. 10 years later	NEW WIRING

WHAT YOU WROTE IN 2001	WHAT IT BECAME	YOUR EXACT WORDS	WHEN IT CAME TO FRUITION	TYPE
A **book downloaded wirelessly** onto a lightweight screen the size of a hardback novel, which **opens to the last page read by you, your spouse or others**, on a **folding screen** p.7	The most specific one. The Kindle delivered wireless downloads, Whispersync resumes at the last page read across devices, and family sharing covers "your spouse or others." Folding screens reached phones in 2019.	*"You pick up your new book (downloaded wirelessly) on a folding screen in the shape of a hardback novel (but lighter in weight) which would open to the last page read by yourself, your spouse or others."*	**November 2007** (Kindle), **February 2009** (Whispersync), **2014** (Kindle Family Library), **September 2019** (Galaxy Fold). 6 to 18 years later	NEW INVENTION
Saving on your **cable bill by getting entertainment through the Internet connection** p.9	Netflix streaming and the whole cord cutting movement.	*"How can they save money on their cable bill by receiving entertainment thru the Internet connection?"*	**January 2007** (Netflix streaming). 6 years later	NEW WIRING
Posting family pictures to a family website and keeping a shared family calendar pp.3, 9	Facebook, shared photo albums, and shared Google and Apple calendars.	*"How about checking all of your email and spending a few minutes checking out the latest pictures posted to the family web site from your family members." / "They can post pictures of the family and update the family calendar and contacts."*	**February 2004** (Facebook), **April 2006** (Google Calendar). 3 to 5 years later	NEW WIRING
Your **cellphone turning up the A/C and lights as you get close** to home p.3	Location based automations in Nest, SmartThings, Google Home and Apple Home. In 2001 most phones couldn't report their location and there were no apps.	*"Your cellphone will turn up the A/C and turn on the lights as you get close to."*	**2011** (Nest thermostat), **2014 to 2016** (phone location automations in Nest, SmartThings and Apple Home). 10 to 15 years later	NEW INVENTION
A **GPS route that bypasses traffic congestion**, with spoken weather and traffic alerts as you get in the car p.8	Live traffic routing in Waze and Google Maps, read aloud through CarPlay and Android Auto.	*"Upon entering your car and the closing of your door, the garage door is lifted, seatbelt reminder and weather alerts are spoken to you along with traffic advisories and a new GPS programmed route, which will bypass any traffic congestion on your way to the office for the day."*	**2007** (Google Maps live traffic), **2009** (Waze), **2014** (CarPlay), **2015** (Android Auto). 6 to 14 years later	NEW WIRING
Ordering pizza, a lawn service and a pool cleaner in **a few clicks**, and **switching contractors with a click** instead of a confrontation pp.3, 9	Angi (formerly Angie's List and HomeAdvisor), Thumbtack, TaskRabbit and DoorDash. Your core business idea became a large category.	*"How about clicking a few check boxes and ordering pizza, a new lawn-mowing service and a pool cleaner in just a few minutes." / "The customer can change the contractor they use with the click of a button instead of a face-to-face confrontation, which so many people feel uncomfortable with."*	**Mid to late 2000s** (Angie's List goes national), **2008** (TaskRabbit and Thumbtack), **2012** (ServiceMagic becomes HomeAdvisor), **2013** (DoorDash). 5 to 12 years later	NEW WIRING

Niche, partial, late or missed 12 PREDICTIONS

WHAT YOU WROTE IN 2001	WHAT IT BECAME	YOUR EXACT WORDS	WHEN IT CAME TO FRUITION	TYPE	VERDICT
Kitchen touch panel with news, calendar, email, security cameras, child monitors and alerts for water heater, plumbing, HVAC and propane p.7	Echo Show and Nest Hub with Ring or Nest camera feeds, plus leak sensors and smart propane monitors. The pieces all exist, but few homes have the full integrated setup.	*"Kitchen touch panel displays CNN, which appears in one corner and your daily appointments in another along with email and voicemail alerts for all phones. Other corners can be security cameras, child monitors, home alert devices (hot water system, plumbing system, hvac system, propane levels), and grocery list."*	**June 2017** (Echo Show), **October 2018** (Google Home Hub). 16 to 17 years later	NEW WIRING	**Came true, partly** The full integrated version is still niche.
One tap "goodnight" and morning sequences for lights, security, thermostat, alarm music and coffee maker pp.3, 7	Alexa and Google routines with smart plugs and thermostats.	*"You load your coffee maker the night before, hop into bed, hit the touch panel to turn all house lights off, except your bedroom, arm the security system and turn on your alarm clock including the setting of your morning wake-up music, the thermostat gets set to the appropriate level and your home/cell phones are set to mute." / "Your alarm clock will start the coffee maker."*	**2017** (Alexa Routines), **2018** (Google Assistant Routines). 16 to 17 years later	NEW WIRING	**Came true**
Smart refrigerator that builds a grocery list, sends it to the store for delivery and flags spoiled food p.3	Samsung Family Hub does much of this with internal cameras and grocery ordering.	*"Refrigerators will be equipped with barcode scanners and will make your grocery list for you and send it to the local store for delivery and they will also alert you to food that is spoiling or outdated."*	**January 2016** (Samsung Family Hub). 15 years later	NEW INVENTION	**Came true, never mainstream** LG showed a $20,000 Internet fridge in 2000 that almost nobody bought. Samsung made it a real product in 2016.
Door locks and security resets automatically when you leave, and the **garage door opens** as you arrive p.8	Smart locks and app controlled garage openers like myQ.	*"As the door shuts, it is locked and security system is reset to proper level for wife's awakening, kitchen lights and coffee maker are shut off. Upon entering your car and the closing of your door, the garage door is lifted"*	**2011** (Chamberlain myQ), **2013** (August Smart Lock). 10 to 12 years later	NEW WIRING	**Came true**

WHAT YOU WROTE IN 2001	WHAT IT BECAME	YOUR EXACT WORDS	WHEN IT CAME TO FRUITION	TYPE	VERDICT
Remote car start with automatic cabin temperature, automatic shutoff, and carbon monoxide protection in the garage p. 8	App based remote start and climate preconditioning, standard on Teslas and many other cars.	*"Your car is started automatically (shuts off in 5 minutes if you have not entered vehicle or if Carbon Monoxide emissions reach unsafe level within garage due to ventilation failure) and car temperature level automatically set depending upon outside temperature."*	**2010** (OnStar RemoteLink app), **2012** (Tesla Model S preconditioning). 9 to 11 years later	NEW INVENTION	**Came true** Except the carbon monoxide sensing, which is still not standard.
Alert that the school bus is coming down your street p. 7	School bus tracking apps used by many districts.	*"A reminder flashes on the screen to make sure that the children have headed out to the curbside as the school bus has just been sensed starting down your street."*	**Mid 2010s** (apps such as Here Comes the Bus). About 13 years later	NEW WIRING	**Came true**
Automatic pet feeding and watering and a **pet door that unlocks** p. 7	Smart feeders and microchip pet doors.	*"the dog and cat bowls are auto filled with food and water" / "Pet door unlocks to allow the dog and cat out."*	**Late 2000s** (microchip pet doors), **mid 2010s** (app controlled feeders). 7 to 14 years later	NEW WIRING	**Came true, niche**
Bills paid automatically on the day they are due, or all at once with a few clicks pp. 3, 7	Autopay became the default for most utilities.	*"Appropriate funds are automatically being transferred to accounts for Utility bills that are due that day." / "How about paying all of those bills at once or at specified times during the month with just a few clicks."*	**Mid 2000s** (mainstream bank bill pay and utility autopay). About 4 to 5 years later	NEW WIRING	**Came true** Online bill pay existed in 2001 but few households used it. Paying automatically on the due date became the norm in the mid 2000s.
Trash can that scans barcodes onto your shopping list, and **reminders on a screen in the bathroom** p. 7	Smart trash cans and smart mirrors have been built.	*"Empty can of shaving cream thrown into wastebasket where it is bar code scanned upon entry and added to computerized shopping list." / "Short list of daily reminders/ appointments is on screen in bathroom or shower for daily planning and clothing decisions while showering and shaving."*	**2016** (GeniCan barcode trash scanner), **2018** (Kohler Verdera smart mirror). 15 to 17 years later	NEW INVENTION	**Came true, novelties**

WHAT YOU WROTE IN 2001	WHAT IT BECAME	YOUR EXACT WORDS	WHEN IT CAME TO FRUITION	TYPE	VERDICT
75% of computer users with high speed access within 2 years p . 3	Home broadband did become the norm.	*"In the next 2 years, 75% of computer-users in the cities and rural communities will have high-speed Internet access."*	**Roughly 2008**, when about three quarters of US Internet users had home broadband (Pew). About 7 years later, not 2	MARKET CALL	**Late** Right direction, about 5 years behind the forecast.
"Contractor brokering has no competition" p . 17	The category became large, led by Angie's List and ServiceMagic (later HomeAdvisor, now Angi).	*"The major service that CES is offering, contractor brokering, has no competition to speak of, as there is currently no other company offering this type of service."*	**1995** (Angie's List, Midwest) and **December 1998** (ServiceMagic, Denver) had started, but neither was a national brand until about **2005 to 2012**.	MARKET CALL	**Partly right** Both were small and regional in 2001. From Florida, the market really did look open.
Ordering all utilities once with a single form when you move p . 3	Moving concierge services handle part of it.	*"Imagine sitting down at the computer the week before your moving into your new home and ordering all of your utilities (i.e. Electricity, Phone, Gas, Rubbish, Cable, Internet, Security, etc) online with just entering your Name, address, phone numbers and SSN just once...no more waiting 20 minutes on the phone for each service."*	**2010** (Updater founded), still only partial today.	NEW WIRING	**Missed so far** Most people still set up utilities one provider at a time.

The GenZPlay plan

WRITTEN JUNE 2013 · SECTION NAMES REFER TO THE ORIGINAL PLAN

GenZPlay built outdoor, interactive fitness and learning apps for kids. Forecasts the plan quotes from other sources are left out; only Stephen's own predictions are included.

Strong hits

9 PREDICTIONS

WHAT YOU WROTE IN 2013	WHAT IT BECAME	YOUR EXACT WORDS	WHEN IT CAME TO FRUITION	TYPE
Interactive stories played outdoors in your backyard or park, using real trees and structures, bringing the virtual world out of the home Executive Summary, Conclusion	Pokémon Go got millions of kids and families walking outside to play. Niantic's Ingress was an invite only beta in June 2013 and didn't launch publicly until December 2013.	*"Imagine an interactive story with many possible outcomes that takes place in your own backyard, or park, using the trees, plants and structures within." / "These methods will bring the virtual world out of the home, which is where children have found their stimulation as of today."*	**July 2016** (Pokémon Go). 3 years later	NEW WIRING
A device on a child's wrist, hip or backpack that becomes second nature, with a motivator built in Executive Summary, Industry Findings	Kids' fitness wearables, activity rings, and Apple Watches for kids who don't have their own iPhone.	*"We want to show the children nature, exploration, mystery and fitness with an electronic device on their wrist, hip or in their backpack. The electronic device should become second nature to all of us" / "An app is an inexpensive way to have a motivator built into the devices we use on a daily basis."*	**April 2015** (Apple Watch activity rings), **2016** (Garmin vivofit jr.), **2018** (Fitbit Ace), **September 2020** (Apple Watch Family Setup). 2 to 7 years later	NEW WIRING
Doctors recognizing the healing power of apps and recommending them to families. The plan's opening also names attention deficit disorder Problem & Opportunity, Marketing	EndeavorRx became the first video game the FDA authorized for doctors to prescribe, to treat ADHD in children ages 8 to 12.	*"Doctors, health teachers and psychotherapists are realizing the healing impact that the new apps are bringing to the world." / "GenZPlay has spoken with many doctors who would be willing to promote the GenZPlay apps as a way of giving extra advice and help for families."*	**June 2020** (EndeavorRx). 7 years later	NEW INVENTION
Body sensors and eHealth as "the next dot.com boom" Problem & Opportunity	Digital health venture funding hit a record $29.1 billion, then fell sharply in 2022 and 2023, following the dot com pattern all the way through.	*"with the current acceleration of body sensor devices and eHealth products, we are seeing the next dot.com boom."*	**2021** (record funding year). 8 years later	MARKET CALL

WHAT YOU WROTE IN 2013	WHAT IT BECAME	YOUR EXACT WORDS	WHEN IT CAME TO FRUITION	TYPE
Teachers turning into "entertainment creators" Problem & Opportunity, Conclusion	Game based classroom tools such as Kahoot, Gimkit and Blooket.	*"As the educational system is now transforming from babbling and confused teachers to entertainment creators, their goal is always the same, and currently is not how to improve communication with their students but how to entertain their students with what they want to learn."*	**Fall 2013** (Kahoot public launch), **2017 to 2018** (Gimkit and Blooket). Months to 5 years later	NEW WIRING
Practically every child able to get a smart device Problem & Opportunity	In 2013, 37% of US teens had a smartphone. By 2018, 95% had one or could use one (Pew).	*"Pricing is stable on the technology market for the first time in the past 50 years, and practically every child can get its hands on a smart device."*	**2018** (Pew: 95% of teens). 5 years later	MARKET CALL
4K televisions as the next big trend for custom installers Custom Integration	4K became the standard TV.	*"Currently, the trend is for security and video systems, and the next trend will most likely be the new 4K televisions"*	**2014** (Netflix 4K streaming), **2018** (over 100 million 4K sets sold worldwide in one year, close to half of all TVs). 1 to 5 years later	MARKET CALL
A Bluetooth tracker on an object, with an app that guides you to where it is Custom Integration	Tile, then Apple AirTag with Precision Finding. Parents now use AirTags for exactly this kind of hunt. Tile opened its crowdfunding the same month the plan was written.	*"Allow the parents to mount a BT tracker to a prize and place it somewhere on their property. The app will setup an adventure and use clues to guide them towards its location."*	**2014** (Tile ships), **April 2021** (AirTag). 1 to 8 years later	NEW WIRING
Kids making YouTube videos about apps and games, and **mom bloggers as a marketing channel** Marketing	Kidfluencers and momfluencers. Ryan's World grew out of a young boy reviewing toys on YouTube.	*"Children can create videos for YouTube of using the app and their suggestions for gameplay and improving." / "There are currently 4 million 'Mommy' bloggers and they are a great resource for marketing as they are always looking for new content for their blogs"*	**March 2015** (Ryan's Toys Review starts), **2018** (Forbes names Ryan the top earning YouTuber). 2 to 5 years later	MARKET CALL

Niche, partial, late or missed 11 PREDICTIONS

WHAT YOU WROTE IN 2013	WHAT IT BECAME	YOUR EXACT WORDS	WHEN IT CAME TO FRUITION	TYPE	VERDICT
Electronic Tag & Chase with Bluetooth LED pods: tap one and the next lights up, and the app times the course Custom Integration	BlazePod: Bluetooth pods you tap as they light up, run from an app that times your reactions.	*"Attach a single or multiple LED's to a BT tracker and mount them to trees, rocks, etc around the yard. The app will create a random courses and time them as they complete it. Children can also tap on the LED/BT tracker and then the next one would light up to extend the course, or follow the lights like a game of interactive TAG."*	**2018** (BlazePod Kickstarter). 5 years later	NEW INVENTION	**Came true, almost exactly** Similar systems were sold only to pro sports teams in 2013. The consumer version arrived in 2018.
Augmented reality Frisbee golf built on Bluetooth hardware Executive Summary	TechDisc, a Bluetooth sensor disc that measures each throw and simulates its flight. The UDisc scoring app already existed.	*"Bluetooth hardware for Virtual 'Augmented Reality' Frisbee Golf"*	**2012** (UDisc app), **early 2020s** (TechDisc). About 8 years later	NEW INVENTION	**Partly came true** Smart discs exist, but full augmented reality course play has not arrived.
Game style nature apps for kids exploring outdoors Competition	Seek by iNaturalist: kids point a phone at plants and animals and earn badges. Merlin added bird identification by sound.	*"This is the beauty of our world outside of the house, but all gaming is currently focused on fantasy worlds explored from the couch. The outdoor world is the best world there is to explore, so lets create 'apps for that'."*	**2018** (Seek), **2021** (Merlin Sound ID). 5 to 8 years later	NEW WIRING	**Came true, niche** Popular with families, small next to mainstream gaming.
A portable encyclopedia with storylines that adapt to how each child learns Executive Summary	AI tutors like Khanmigo, plus ChatGPT in every pocket.	*"allow our children to hold a portable encyclopedia of fun and learning. We also need to build storylines and content that will adapt to the children's learning preferences and excite them to be active."*	**November 2022** (ChatGPT), **March 2023** (Khanmigo). About 10 years later	NEW INVENTION	**Coming true now** Adaptive learning software existed in 2013, but content that truly adapts only arrived with AI.
A password protected reward system that parents or older siblings control Problem & Opportunity	Chore and allowance apps like BusyKid and Greenlight, plus Apple Screen Time and Google Family Link.	*"a password protected rewards system administered by parents or older siblings."*	**2016 to 2018** (BusyKid, Greenlight, Family Link, Screen Time). 3 to 5 years later	NEW WIRING	**Came true** ChoreMonster, which the plan lists, already did this in 2013.
A middle school education through massive multiplayer learning apps Problem & Opportunity	Minecraft Education in classrooms, full time virtual schools, and the switch to remote learning in 2020.	*"Will a 'Middle School' degree be available, from the use of Massive Multiplayer Educational apps, before you graduate on to an actual High School? Many menial courses/classes can be bypassed or completed via an app model."*	**November 2016** (Minecraft Education Edition), **2020** (widespread remote school). 3 to 7 years later	MARKET CALL	**Partly came true** Game based learning is common, but nobody earns a diploma from a game.

WHAT YOU WROTE IN 2013	WHAT IT BECAME	YOUR EXACT WORDS	WHEN IT CAME TO FRUITION	TYPE	VERDICT
Technology to help with children's diabetes Industry Findings	Dexcom Share lets parents see a child's glucose readings on their own phones.	*"If we combine technology with kinesiology programs and other health related issues, such as children's diabetes, we can teach, entertain and coach our kids to be the person they want to be."*	**October 2014** (Dexcom Share cleared by the FDA). 1 year later	NEW INVENTION	**Came true, partly** On the monitoring side, not as teaching and coaching games.
Smart backyard features: LED landscape lighting, yard speakers, solar charging, playhouses with electronic whiteboards Custom Integration	App controlled outdoor lighting is common. Outdoor speakers were already standard work for installers.	*"LED's can double as landscape accent lighting" / "Speakers in the yard for music and gameplay" / "Solar charging station for electronics" / "Integrated playhouses with electronic white boards"*	**2018** (Philips Hue Outdoor). 5 years later	NEW WIRING	**Mixed** Smart outdoor lighting came true. Electronic whiteboard playhouses never caught on.
About a year left before the industry peaked, with the technology **within 12 months** Advantage	The big moment for outdoor games was Pokémon Go in 2016. The consumer pod hardware came in 2018.	*"This window of opportunity is just opening and we have about a year to perfect and market our products before the industry pinnacle." / "Creating this technology is easy, as the hardware and software systems are already in place or are within 12 months of fruition."*	**2016 to 2018**. 3 to 5 years later	MARKET CALL	**Late** Right direction, but 3 to 5 years instead of 1.
Patents and copyright wouldn't hold back apps Business Model	App developers faced patent suits and copycat crackdowns.	*"The only difference is, in the past, you had to create something absolutely unique due to patents and copyright infringements. The app market is nothing like this."*	Already happening: **2011** (Lodsys sues app developers), **August 2012** (Apple v. Samsung verdict).	MARKET CALL	**Missed** This was already untrue when the plan was written.
Parents' worries about kids and technology would fade as Gen Y became parents Problem & Opportunity	It went the other way: The Anxious Generation, school phone bans in many US states, and Australia's social media ban for children under 16.	*"The Baby-Boomer parents, who didn't quite understand how the world changed so fast with the introduction of TV's and computers, instigated the recent apprehension of technology use with our children... We now know that technology is just a more entertaining looking glass into a world of education and understanding, so we need to take back time and give children quality access to technology."*	**2024** (The Anxious Generation, state phone bans), **December 2025** (Australia's ban takes effect).	MARKET CALL	**Missed** The worry grew instead of fading. The plan's own warning about devices turning kids into "little zombies" is the view that won out.

What the two plans show

New wiring arrives first.

Ideas that combined parts already on the market usually showed up within about 3 to 10 years, and sometimes within months.

New inventions take longer.

Reading position sync took 8 years, phone location automations 10 to 15, and the folding screen 18.

Right direction, early timing.

The market calls were mostly correct about where things were heading and early about when. A good rule of thumb: when something looks one year away, plan for five.

A twelve year rhythm.

2001, 2013 and 2025 each came a few years after a new platform arrived: the web, the smartphone, and AI. The idea running through all three is the same: technology should give people their time back and stay accountable to them.

SOURCES FOR DATES AND FACTS

- Angie's List, Encyclopedia of Indianapolis
- How Angie's List Started (Powderkeg interview with Angie Hicks)
- HomeAdvisor (Wikipedia)
- IAC Acquires ServiceMagic (Feld Thoughts)
- Seek app lets kids enjoy nature (Forbes)
- BlazePod review (Gadgets & Wearables)
- UDisc (Wikipedia)
- From trial and error to TechDisc (UDisc blog)
- Annual 4K TV sales to surpass 100 million (FlatpanelsHD)
- Digital health funding reaches record $29.1B in 2021 (Healthcare Dive)

Other launch dates are drawn from widely reported public records. Compiled September 2026.